

0 1 . 1

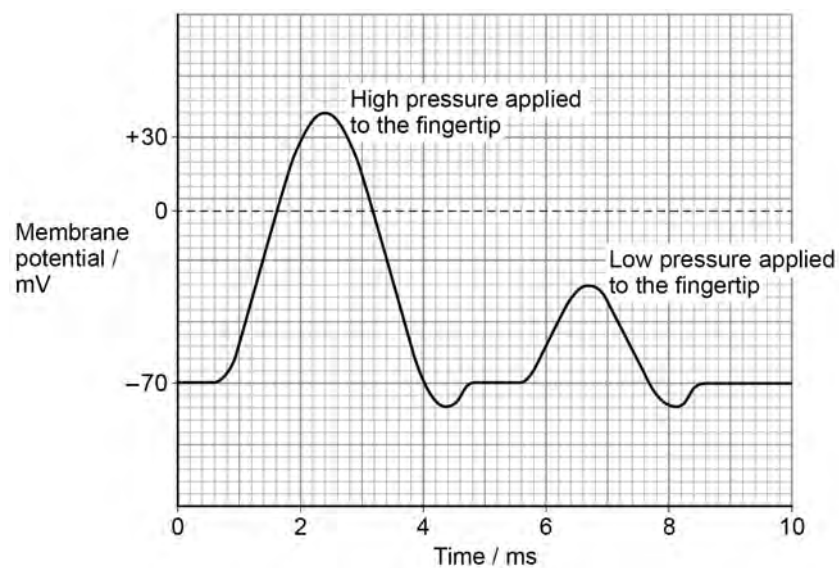
Describe how stimulation of a Pacinian corpuscle produces a generator potential.

[3 marks]

Scientists investigated the stimulation of a Pacinian corpuscle in the skin of a fingertip. The scientists applied two different pressures to the fingertip and recorded the changes in membrane potential of the Pacinian corpuscle's sensory neurone.

Figure 1 shows the scientists' results.

Figure 1



0 1 . 2

Use **Figure 1** to describe what is meant by the all-or-nothing principle.

[2 marks]

0 1 . 3

On **Figure 1**, from 0.6 ms to 4.0 ms, no new generator potential could be produced. What is this time period called?

[1 mark]

Turn over for the next question

